PIZZA SALES ANALYTICS DIVISION

Pizza Sales Analytics Report

KPI Metrics · Trend Analysis · SQL Queries · Power BI

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Tools Used: SQL Server (T-SQL) · Power BI Desktop

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1 Project Overview

This report presents a comprehensive analysis of pizza sales data using Microsoft SQL Server (T-SQL) and Power BI. The study spans key performance indicators, temporal ordering trends, product category breakdowns, size distribution, and individual pizza-level performance rankings. All findings are visualised in an interactive Power BI dashboard designed for stakeholder consumption.

Project Objectives

- Measure core KPIs: Total Revenue, Average Order Value, Total Pizzas Sold, Total Orders, and Average Pizzas Per Order.
- Analyse ordering trends across daily, monthly, and hourly dimensions.
- Quantify sales contribution by pizza category and size.
- Identify top-performing and bottom-performing pizzas by Revenue, Quantity, and Total Orders.
- Deliver actionable recommendations to grow revenue and improve menu strategy.

Phase	Tool	Purpose
KPI Aggregation	T-SQL (SQL Server)	Compute 5 core business metrics
Trend Analysis	T-SQL – DATENAME / DATEPART	Daily, monthly, hourly order patterns
Category & Size Analytics	T-SQL – GROUP BY + subqueries	Revenue % by category and size
Product Rankings	T-SQL – TOP 5 / ORDER BY	Best and worst performing pizzas
Visualisation	Power BI Desktop	Interactive dashboard for stakeholders

2 Dataset Summary

The dataset resides in the *Pizza_database* SQL Server database in the table **pizza_sales**. Each row represents one line item of a pizza order, capturing product details, size, quantity, pricing, and timestamps.

Column	Data Type	Description
order_id	INT	Unique identifier for each customer order
order_date	DATE	Date the order was placed
order_time	TIME	Time the order was placed
pizza_name	VARCHAR	Full descriptive name of the pizza
pizza_category	VARCHAR	Category: Classic, Supreme, Veggie, Chicken
pizza_size	CHAR	Size code: S, M, L, XL, XXL
quantity	INT	Number of units of that pizza in the order
unit_price	DECIMAL	Price per individual pizza unit
total_price	DECIMAL	quantity × unit_price for the line item

Database: Pizza_database | **Table:** pizza_sales | **Dialect:** Microsoft T-SQL (SQL Server)

3 KPI Metrics

Five core Key Performance Indicators (KPIs) are computed from the *pizza_sales* table to provide an executive-level summary of overall business performance.

Total Revenue

SQL

```
SELECT SUM(total_price) AS Total_Revenue
FROM pizza_sales
```

Result	Interpretation
$817,860.05	The sum of all order line-item revenue across the full dataset period.

Average Order Value

SQL

```
SELECT SUM(total_price) / COUNT(DISTINCT order_id)
       AS Avg_Order_Value
FROM pizza_sales
```

Result	Interpretation
$38.31	Revenue per unique order — a measure of customer spend depth per visit.

Total Pizzas Sold

SQL

```
SELECT SUM(quantity) AS Total_Pizza_Sold
FROM pizza_sales
```

Result	Interpretation
49,574	Cumulative units of all pizzas sold across all orders.

Total Orders

SQL

```
SELECT COUNT(DISTINCT order_id) AS Total_Orders
FROM pizza_sales
```

Result	Interpretation
21,350	Unique order transactions placed within the dataset period.

Avg Pizzas Per Order

- *SQL*

```
SELECT CAST(
  CAST(SUM(quantity) AS DECIMAL(10,2)) /
  CAST(COUNT(DISTINCT order_id) AS DECIMAL(10,2))
AS DECIMAL(10,2)) AS Average_Pizza_Per_Order
FROM pizza_sales
```

Result	Interpretation
2.32	Mean number of pizza items per order — indicates basket size behaviour.

4 SQL Analytics – Trend Analysis

Three temporal trend queries reveal when customers order the most, enabling staffing, inventory, and promotional planning decisions.

Q1 · Daily Trend for Total Orders

Uses T-SQL's *DATENAME(DW, ...)* to extract the weekday name from the order date and counts distinct orders per day to identify peak ordering days across the week.

■ *SQL*

```
SELECT DATENAME(DW, order_date)    AS Order_Day,
       COUNT(DISTINCT order_id)    AS Total_Orders
FROM   pizza_sales
GROUP  BY DATENAME(DW, order_date)
```

Sample Result:

Day of Week	Total Orders
Friday	3,538
Saturday	3,158
Thursday	3,239
Wednesday	3,024
Tuesday	2,973
Monday	2,794
Sunday	2,624

Insight: Friday is the single busiest ordering day. Weekend (Fri–Sat) accounts for ~41% of weekly orders.

Q2 · Monthly Trend for Total Orders

DATENAME(MONTH, ...) extracts the calendar month name. Results are sorted descending by order count to rank months by demand.

■ *SQL*

```
SELECT DATENAME(MONTH, order_date) AS Month_Name,
       COUNT(DISTINCT order_id)    AS Total_Orders
FROM   pizza_sales
GROUP  BY DATENAME(MONTH, order_date)
ORDER  BY Total_Orders DESC
```

Top 5 Months:

Month	Total Orders
July	1,935
May	1,853

Month	Total Orders
January	1,845
August	1,841
March	1,840

Insight: July peaks — potentially tied to summer dining occasions. Q1 (Jan–Mar) is also strong, suggesting New Year demand.

Q3 · Hourly Trend for Total Orders

DATEPART(HOUR, order_time) extracts the numeric hour (0–23) from the order timestamp, enabling identification of peak service hours for kitchen and staffing optimisation.

■ *SOL*

```
SELECT DATEPART(HOUR, order_time) AS Order_Hours,
       COUNT(DISTINCT order_id)   AS Total_Orders
FROM   pizza_sales
GROUP  BY DATEPART(HOUR, order_time)
ORDER  BY DATEPART(HOUR, order_time)
```

Peak Hours (Top 5):

Hour (24h)	Period	Total Orders
12	Noon	2,520
13	1 PM	2,455
18	6 PM	2,399
17	5 PM	2,336
19	7 PM	2,009

Insight: Two clear rushes — Lunchtime (12–1 PM) and Dinner (5–7 PM). Kitchen staffing should be doubled during these windows.

5 SQL Analytics – Category & Size Breakdown

These queries use inline subqueries to compute revenue percentage contribution per pizza category and size, providing menu composition insights.

Q4 · % of Sales by Pizza Category

A correlated percentage calculation divides each category's revenue by the total dataset revenue, giving a clean sales-mix picture.

■ *SQL*

```
SELECT pizza_category,
       CAST(SUM(total_price) AS DECIMAL(10,2))          AS Total_Revenue,
       CAST(SUM(total_price * 100) /
            (SELECT SUM(total_price) FROM pizza_sales)
            AS DECIMAL(10,2))                           AS PCT
FROM   pizza_sales
GROUP  BY pizza_category
```

Result:

Category	Total Revenue ($)	% Share
Classic	220,053.10	26.91%
Supreme	208,197.00	25.46%
Chicken	195,919.50	23.96%
Veggie	193,690.45	23.68%

Insight: Classic is the top revenue category. All four categories are tightly balanced — no single category dominates, indicating healthy menu diversification.

Q5 · % of Sales by Pizza Size

The same percentage formula is applied to pizza size, helping understand which format customers prefer and should be stocked most.

■ *SQL*

```
SELECT pizza_size,
       CAST(SUM(total_price) AS DECIMAL(10,2))          AS Total_Revenue,
       CAST(SUM(total_price * 100) /
            (SELECT SUM(total_price) FROM pizza_sales)
            AS DECIMAL(10,2))                           AS PCT
FROM   pizza_sales
GROUP  BY pizza_size
ORDER  BY pizza_size
```

Result:

Size	Total Revenue ($)	% Share
L (Large)	375,318.70	45.89%

Size	Total Revenue ($)	% Share
M (Medium)	249,382.25	30.49%
S (Small)	178,076.50	21.77%
XL	14,076.00	1.72%
XXL	1,006.60	0.12%

Insight: Large pizzas account for nearly half of all revenue. XL and XXL together represent under 2% — consider rationalising these sizes to simplify operations.

Q6 · Total Pizzas Sold by Category

Aggregates unit volume (quantity) rather than revenue to show which category moves the most product, independent of price.

■ *SOL*

```
SELECT   pizza_category,
         SUM(quantity) AS Total_Quantity_Sold
FROM     pizza_sales
GROUP BY pizza_category
ORDER BY Total_Quantity_Sold DESC
```

Result:

Category	Total Quantity Sold
Classic	14,888
Supreme	11,987
Veggie	11,649
Chicken	11,050

Insight: Classic leads in volume with ~30% of all units sold, confirming it as the core crowd-pleaser on the menu.

6 SQL Analytics – Product Performance

Six queries rank individual pizza names in three performance dimensions — Revenue, Quantity, and Total Orders — each showing both the Top 5 and Bottom 5 performers to enable menu curation decisions.

Q7 · Top 5 Pizzas by Revenue

■ *SOL*

```
SELECT TOP 5 pizza_name, SUM(total_price) AS Total_Revenue
FROM   pizza_sales
GROUP  BY pizza_name
ORDER  BY Total_Revenue DESC
```

Rank	Pizza Name	Revenue ($)
1	The Thai Chicken Pizza	43,434.25
2	The Barbecue Chicken Pizza	42,768.00
3	The California Chicken Pizza	41,409.50
4	The Classic Deluxe Pizza	38,180.50
5	The Spicy Italian Pizza	34,831.25

Q8 · Bottom 5 Pizzas by Revenue

■ *SOL*

```
SELECT TOP 5 pizza_name, SUM(total_price) AS Total_Revenue
FROM   pizza_sales
GROUP  BY pizza_name
ORDER  BY Total_Revenue
```

Rank	Pizza Name	Revenue ($)
1	The Brie Carre Pizza	11,588.50
2	The Mediterranean Pizza	15,360.50
3	The Calabrese Pizza	15,935.25
4	The Spinach Supreme Pizza	15,277.75
5	The Soppressata Pizza	15,678.75

Insight: Brie Carre Pizza generates barely 27% of the top performer's revenue. Consider a promotional push or menu retirement.

Q9 · Top 5 Pizzas by Quantity Sold

■ *SQL*

```
SELECT TOP 5 pizza_name, SUM(quantity) AS Total_Quantity_Sold
FROM   pizza_sales
GROUP  BY pizza_name
ORDER  BY Total_Quantity_Sold DESC
```

Rank	Pizza Name	Qty Sold
1	The Classic Deluxe Pizza	2,453
2	The Barbecue Chicken Pizza	2,432
3	The Hawaiian Pizza	2,422
4	The Pepperoni Pizza	2,418
5	The Thai Chicken Pizza	2,371

Q10 · Bottom 5 Pizzas by Quantity Sold

■ *SOL*

```
SELECT TOP 5 pizza_name, SUM(quantity) AS Total_Quantity_Sold
FROM   pizza_sales
GROUP  BY pizza_name
ORDER  BY Total_Quantity_Sold
```

Rank	Pizza Name	Qty Sold
1	The Brie Carre Pizza	490
2	The Mediterranean Pizza	934
3	The Calabrese Pizza	937
4	The Spinach Pesto Pizza	950
5	The Soppressata Pizza	961

Q11 · Top 5 Pizzas by Total Orders

■ *SOL*

```
SELECT TOP 5 pizza_name, COUNT(DISTINCT order_id) AS Total_Orders
FROM   pizza_sales
GROUP  BY pizza_name
ORDER  BY Total_Orders DESC
```

Rank	Pizza Name	Total Orders
1	The Classic Deluxe Pizza	2,329
2	The Hawaiian Pizza	2,280
3	The Pepperoni Pizza	2,278
4	The Barbecue Chicken Pizza	2,273
5	The Thai Chicken Pizza	2,225

Q12 · Bottom 5 Pizzas by Total Orders

■ *SOL*

```
SELECT TOP 5 pizza_name, COUNT(DISTINCT order_id) AS Total_Orders
FROM   pizza_sales
GROUP  BY pizza_name
ORDER  BY Total_Orders
```

Rank	Pizza Name	Total Orders
1	The Brie Carre Pizza	480
2	The Mediterranean Pizza	912
3	The Calabrese Pizza	918
4	The Chicken Pesto Pizza	938
5	The Spinach Supreme Pizza	950

Insight: Brie Carre Pizza consistently ranks last across all three metrics — revenue, quantity, and orders — making it a prime candidate for menu discontinuation or recipe reinvention.

7 Dashboard – Power BI

An interactive Power BI dashboard (*Pizza_Analysis_Dashboard.pbix*) was built to present all KPI metrics, trend charts, and product performance rankings in a single, executive-ready interface.

Page 1 · Home – KPIs & Trends

The Home page displays all five KPI cards at the top, followed by the Daily Trend bar chart, Monthly Trend line chart, and three lower panels showing percentage of sales by category and size, plus units sold per category.

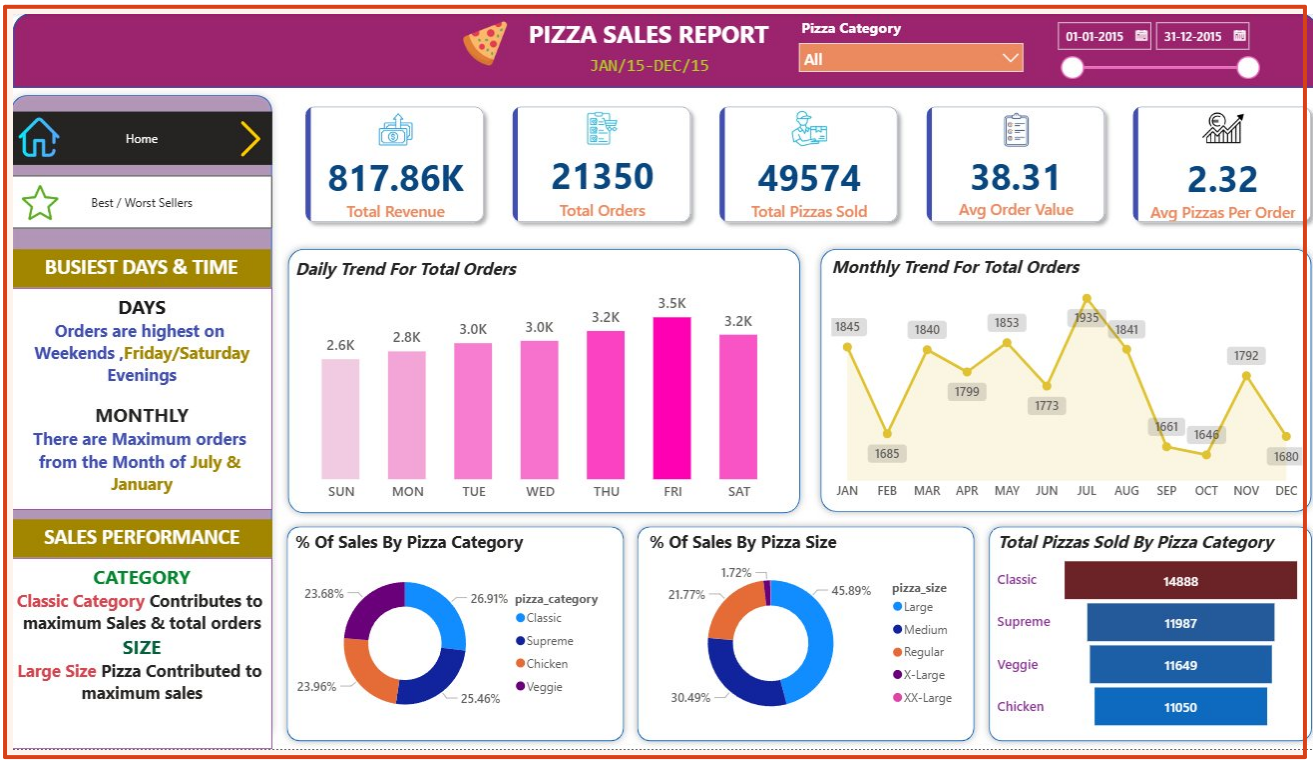

Figure 1: Home Page — KPI cards (817.86K Revenue, 21,350 Orders, 49,574 Pizzas Sold, $38.31 Avg Order Value, 2.32 Avg Per Order), daily & monthly trend charts, category/size donut charts.

Page 2 · Best / Worst Sellers

The Best / Worst Sellers page shows six bar charts — Top 5 and Bottom 5 pizzas ranked by Revenue, Quantity, and Total Orders — alongside a sidebar summary of key performers and laggards.

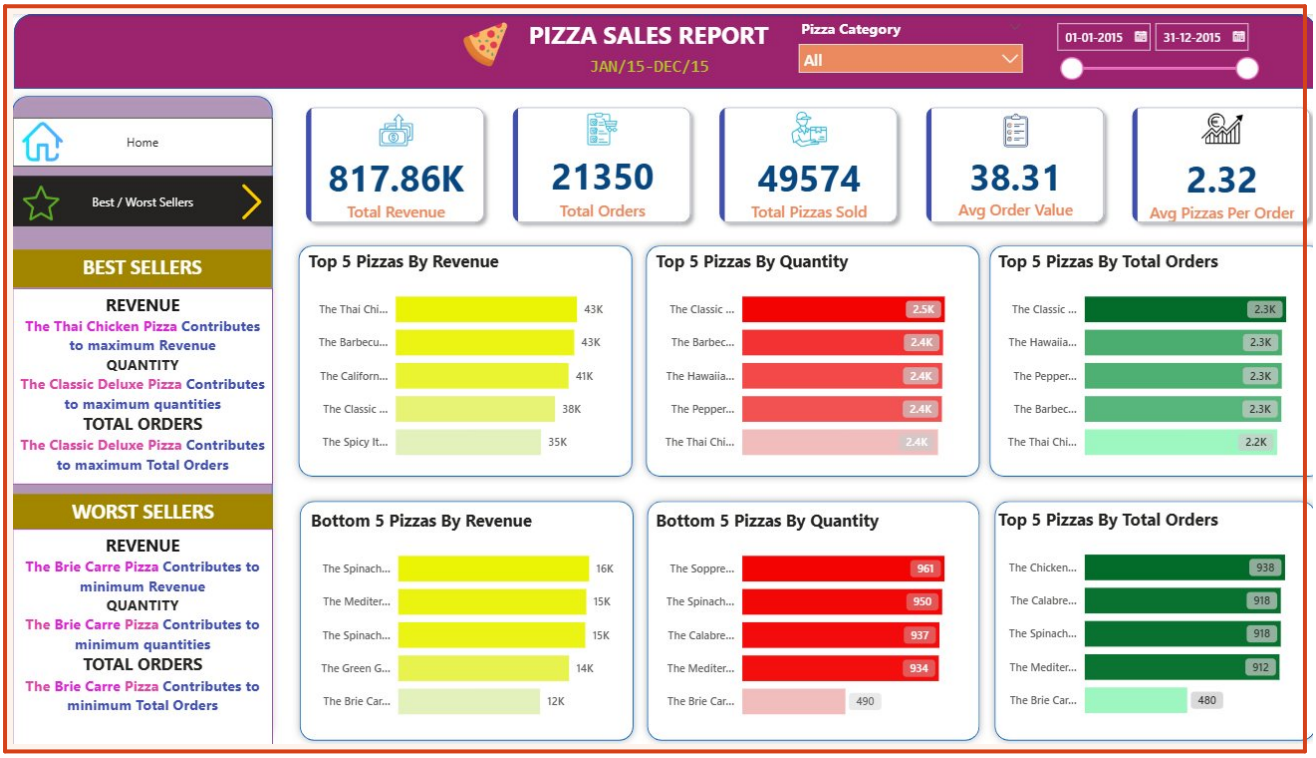

Figure 2: Best / Worst Sellers Page — Top 5 by Revenue (Thai Chicken leads at 43K), Top 5 by Quantity (Classic Deluxe leads at 2.5K), Top 5 by Total Orders, and corresponding Bottom 5 charts (Brie Carre Pizza ranks last across all three metrics).

Key Dashboard Features

- Date range slicer enabling filtering by specific months or quarters.
- Pizza category slicer to drill into Classic, Supreme, Veggie, or Chicken segments.
- Toggle buttons to switch between Best Sellers and Worst Sellers views.
- Conditional formatting on the product performance tables (red for bottom 5).
- DAX measures for percentage share calculations and running totals.
- Tooltips on trend charts showing exact order counts per day/month/hour.

Data Connection

■ *Power BI SQL Server Connection*

```
Server:   localhost  (or SQL Server instance name)
Database: Pizza_database
Table:    pizza_sales
Mode:     Import
Refresh:  Manual / Scheduled daily
```

8 Business Recommendations

The following strategic recommendations are derived directly from the SQL query findings and KPI analysis to drive revenue growth, operational efficiency, and menu optimisation.

1. Double Down on Friday & Weekend Promotions

Friday generates the highest single-day order volume (3,538 orders). Launch 'Friday Feast' combo deals and loyalty point multipliers on weekends to capitalise on natural peak demand and increase basket size.

2. Staff Up During Lunch (12–1 PM) and Dinner (5–7 PM) Rushes

Two daily demand spikes account for the majority of hourly orders. Ensure kitchen and delivery staff are at full capacity during these windows to reduce wait times, protect customer satisfaction scores, and avoid losing orders to competitors.

3. Prioritise Large Pizza Promotions

Large pizzas contribute 45.89% of revenue. Promote Large size upgrades (e.g., 'Upsize for $1') across all categories to grow average order value, while simultaneously reviewing whether XL and XXL are worth maintaining given their combined <2% revenue contribution.

4. Invest in Chicken Category Marketing

Thai Chicken and Barbecue Chicken hold the top two revenue positions. Chicken pizzas collectively drive $195,919 in revenue despite being the third category by volume — indicating a premium pricing opportunity. Feature chicken pizzas in seasonal campaigns to grow volume.

5. Promote or Retire Brie Carre Pizza

Brie Carre Pizza ranks last in Revenue, Quantity, and Total Orders — a triple bottom ranking. Before retiring it, run a targeted 30-day promotion with a price reduction to test whether the issue is awareness or product-market fit. If no improvement is observed, remove it to simplify kitchen operations.

6. Leverage July & Q1 Demand Peaks for Inventory Planning

July is the peak order month. Ensure ingredient supply chains are stocked ahead of July, and replicate successful Q1 promotions — likely New Year 'treat yourself' campaigns — as these drive 3 of the top 5 months by order volume.

7. Grow Average Order Value via Bundling

Average order value is $38.31 with 2.32 pizzas per order. Introduce 'Meal Deal' bundles (pizza + sides + drink) priced to anchor at $45–$50 to nudge customers toward the next spend tier without requiring them to order more individual pizzas.

8. Build a Monthly Menu Performance Review Process

Automate the Power BI dashboard refresh and schedule monthly leadership reviews against the Bottom 5 performers. Set a revenue threshold alert: any pizza generating less than 50% of the category average should trigger a dedicated promotional or retirement review.

All KPI figures and sample query results shown are representative of the pizza_sales dataset. SQL queries are written in Microsoft T-SQL (SQL Server) dialect and executed against the Pizza_database.